

Artur lasenovets, M.S. candidate

Present Position	Graduate Student Researcher Chongqing University of Posts and Telecommunications	Email: L202420002@stu.cqupt.edu.cn Publication list: ORCID
Research Interests	My interests lie in applied cryptography and privacy-enhancing technologies , with a focus on homomorphic encryption , private information retrieval , searchable encryption , and privacy-preserving systems. I am particularly interested in practical cryptographic protocols for private data search and record retrieval.	
Additional Information	Research profiles: Portfolio • LinkedIn References: Available upon request.	
Education	Chongqing University of Posts and Telecommunications , Chongqing, China <i>School of Cyberspace Security and Information Law</i> ◇ M.S. Network & Information Security Sep. 2024–Jul. 2027 (<i>expected</i>) ◇ Supervisor: Prof. Tang Fei ◇ GPA: 3.98/4.00 Ukhta State Technical University, Ukhta, Russia ◇ B.S. Information Systems and Technologies Mar. 2022–Jun. 2024 ◇ GPA: 4.46/5.00 St. Petersburg State University of Telecommunications, St. Petersburg, Russia ◇ B.S. studies in Information Systems and Technologies Sep. 2020–Mar. 2022	
Publications	Publications and Preprints Artur lasenovets , Fei Tang, H. Zhu, P. Wang, and L. Liu. “ Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval ,” arXiv:2511.02656, 2025. Manuscript under review. H. Zhu, F. Tang, W. Qin, J. Shan, P. Wang, and Artur lasenovets . “ EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption ,” <i>Journal of King Saud University – Computer and Information Sciences</i> , 2025. Artur lasenovets and Fei Tang. “ MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework ,” <i>International Research Journal</i> , no. 1 (163), 2026. Manuscript in Preparation Piyumi M. S. Rajapaksha*, Artur lasenovets *, Yinxue Yi, and Fei Tang. “TWAPPA-FRL: Trust-Weighted Privacy-Preserving Aggregation for Federated Reinforcement Learning in Collaborative Intrusion Detection,” manuscript in preparation, 2026. *Equal contribution.	
Awards & Achievements	Chinese Government Scholarship for Graduate Studies 2024–2027 Full scholarship for the M.S. programme at CQUPT. Winner, YouKnow EdTech Project Competition 2023 Won the EdTech project competition organized by the MIPT Phystech School, the Fund for Development of Phystech Schools, and Startech.Base. First Degree Diploma, Severgeoecontech 2022 Awarded first place in the Modern Information Technologies section of the 23rd International Youth Scientific Conference. Sber Student Accelerator Regional Demo Day 2023 Completed the Sber Student Accelerator with the AcademAI team and participated in the North-Western regional demo day. Supporting certificates	

Research Experience	Graduate Student Researcher @ Cryptography and Applications Sep. 2024–Present Research Group, CQUPT	
	Private Information Retrieval for Hyperledger Fabric - Designed and implemented a BGV-based computational PIR protocol in Go using Lattigo, integrating private world-state reads directly into Hyperledger Fabric chaincode. - Defined the honest-but-curious (HBC) threat model and analysed query privacy under the IND-CPA security of BGV, including repeated-query security and observable leakage. - Built the experimental workflow and evaluated correctness, latency, communication, storage, and deployment overhead across several cryptographic parameter sets.	
	Privacy-Preserving Aggregation for Federated Reinforcement Learning - Co-developed TWAPPA-FRL, combining trust-weighted aggregation (TWA) with CKKS-based privacy-preserving aggregation (PPA) for collaborative intrusion detection, as an equal-contribution author. - Combined trust-aware client weighting with CKKS-based encrypted aggregation of federated model updates and aggregate-only decryption by a separate non-training key authority. - Developed the threat model and security analysis, including individual-update confidentiality, explicit leakage, and the assumptions and limitations of the PPA-enabled paths.	
	Verifiable Multi-Keyword Searchable Encryption - Contributed to EVMK-SSE, an efficient multi-keyword searchable encryption scheme based on an OT-based client-independent relaxed OPRF. - Reviewed the security analysis covering query privacy, explicit search and access-pattern leakage, and result verifiability against a malicious cloud server.	
Industry Experience	Security Intern @ Positive Technologies Aug. 2024–Feb. 2025	
	- Developed tested Python security tooling for the MaxPatrol EDR team to validate Windows file paths, network configurations, and endpoint-control policies. - Built an Airflow-based malware-processing pipeline that collected samples from VX-Underground, stored artifacts in S3/MinIO, and performed YARA-based scanning. - Investigated exploitation and post-exploitation activity involving CVE-2020-5902 and CVE-2023-38831 through EVTX analysis, PowerShell deobfuscation, IoC extraction, and MITRE ATT&CK mapping. - Extended this threat-intelligence experience into the first-author MFCTIIF research project on multi-feed CTI integration.	
	Software Engineer @ AcademAI Aug. 2023–Aug. 2024	
	- Co-developed an end-to-end AI platform for generating structured and personalized educational courses from user-defined topics and learning objectives. - Implemented a Python REST API and LangChain-based generation pipeline using structured prompting and asynchronous LLM orchestration. - Built the frontend and application layer using Next.js, TypeScript, Prisma, and Auth.js, including authentication and persistent course data. - Containerized and deployed the platform using Docker and GitHub Actions.	
Skills	◇ Go, Python, TypeScript: proficient ◇ Homomorphic encryption: BGV, BFV, CKKS ◇ HE libraries: Lattigo, Microsoft SEAL, OpenFHE, TenSEAL ◇ Security engineering: YARA, Chainsaw, malware and log analysis	◇ C++, Rust: familiar ◇ Private search: PIR/CPIR, searchable encryption ◇ Systems: Hyperledger Fabric, Linux, Docker, Git ◇ Languages: Russian (native), English (C1), Chinese (HSK 3)